

THE GASTROINTESTINAL BRANCH OF THE SIALIC-GLYCAN INFLAMMATORY CASCADE:

A Unified Framework for Celiac Disease, Non-Celiac Gluten Sensitivity, Irritable Bowel Syndrome, Dermatitis Herpetiformis, and the Role of Neu5Gc Xenosialitis in Gastrointestinal Inflammatory Burden

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Abstract

Background: The gastrointestinal tract is the first and most direct site of encounter between dietary xenoantigens and the human immune system. Celiac disease, non-celiac gluten sensitivity (NCGS), irritable bowel syndrome (IBS), and dermatitis herpetiformis (DH) represent a spectrum of conditions in which the intestinal immune response to wheat-derived antigens plays a central pathological role. Despite substantial research investment, the mechanistic relationships between these conditions — and the role of a concurrent dietary antigen, Neu5Gc from bovine red meat and dairy, in modulating their severity — have not been examined within a unified framework.

Hypothesis: We propose that the gastrointestinal branch of the Sialic-Glycan Inflammatory Cascade (SGIC) operates through three convergent mechanisms: (1) ATI-mediated TLR4 innate immune activation at the intestinal epithelium and in mesenteric lymph node myeloid cells, producing the gastrointestinal innate inflammatory state that characterises NCGS and IBS; (2) gliadin-mediated zonulin-driven tight junction disruption, enabling both HLA-DQ2/DQ8-restricted adaptive immune responses in celiac disease and the non-celiac systemic translocation of dietary antigens including Neu5Gc that amplifies the SGIC downstream; and (3) Neu5Gc xenosialitis at the intestinal epithelium, which pre-conditions the mucosal immune environment, damages tight junction proteins independently of gliadin, and substantially amplifies the inflammatory response to wheat-derived antigens through concurrent NF-κB activation.

The tissue-resident memory T cell (TRM cell) model established in the Hamer Loop Manuscript Series applies with particular force to the gastrointestinal context: repeated dietary antigen exposure seeds the intestinal lamina propria with antigen-specific TRM cells capable of producing rapid-onset gastrointestinal symptoms within minutes of re-exposure — a timeline inconsistent with classical food intolerance kinetics but directly consistent with the Soup Flare observation documented in the Arthritis/Soup Flare sentinel case report.

Significance: The conditions addressed in this paper collectively affect hundreds of millions of people globally and are characterised by variable diagnostic clarity, significant patient burden, and management approaches that focus on gluten alone without addressing the Neu5Gc component that may be sustaining and amplifying the gastrointestinal inflammatory state. This

paper provides a mechanistic framework for the full gastrointestinal branch and a set of testable predictions for formal investigation.

Keywords: *celiac disease; non-celiac gluten sensitivity; IBS; irritable bowel syndrome; dermatitis herpetiformis; Neu5Gc; xenosialitis; ATIs; gliadins; TLR4; zonulin; intestinal permeability; TRM cells; gut-skin axis; HLA-DQ2; HLA-DQ8; SGIC; dietary antigen*

1. Introduction: The Gut as First Contact

The intestinal epithelium is the largest immunological interface in the human body. Covering a surface area of approximately 30 square metres, it mediates the constant negotiation between the external antigenic environment — food, microbes, and their products — and the internal immune system that must decide what is friend, what is foe, and what requires active response. It is also, within the SGIC framework, the site where both primary dietary xenoantigens — wheat-derived ATIs and gliadins, and bovine-derived Neu5Gc — make first contact with the human immune system.

The gastrointestinal consequences of this contact have been studied primarily through the lens of wheat and gluten, with celiac disease receiving the most mechanistic attention and NCGS and IBS occupying progressively less well-characterised territory. What has not been examined is whether a second dietary antigen — Neu5Gc, incorporated into gut epithelial glycoproteins and engaged by anti-Neu5Gc antibodies at the mucosal surface — constitutes an independent and synergistic contributor to the gastrointestinal inflammatory burden in these conditions.

This paper proposes that it does. It further proposes that the full clinical spectrum from IBS through NCGS to celiac disease and DH represents a continuum of gastrointestinal antigen reactivity in which the combined burden of wheat-derived and bovine-derived xenoantigens determines the severity and clinical expression of the condition — and in which management strategies that address only the wheat-derived component while leaving the Neu5Gc component intact are achieving partial rather than complete resolution by design.

2. The Gastrointestinal Spectrum: From IBS to Celiac Disease

2.1 Irritable Bowel Syndrome: The Undiagnosed End of the Spectrum

Irritable bowel syndrome affects an estimated 10–15% of the global population and is characterised by recurrent abdominal pain, bloating, and altered bowel habit — diarrhoea, constipation, or mixed pattern — in the absence of identifiable structural or biochemical abnormality on standard investigation. It is among the most common reasons for gastroenterological consultation worldwide and among the most frustrating for both patients and clinicians, given the absence of a clear mechanistic explanation and the limited efficacy of available treatments.

Within the SGIC framework, IBS — particularly the diarrhoea-predominant subtype (IBS-D) — is proposed to represent the lowest-threshold end of the gastrointestinal antigen reactivity spectrum: the clinical expression of ATI-driven TLR4 intestinal innate activation and Neu5Gc xenosialitis at the gut epithelium in individuals whose immune reactivity is sufficient to produce functional gastrointestinal disturbance but who do not mount the HLA-DQ2/DQ8-restricted adaptive immune response that characterises celiac disease or the IgA-mediated response that characterises DH.

The rapid-onset gastrointestinal symptoms — bloating and diarrhoea within 40 minutes of trigger food ingestion, as documented in the Soup Flare sentinel observation — are inconsistent with the 1–4 hour latency predicted by classical food intolerance models but directly consistent with TRM cell-mediated rapid immune activation in a sensitised gut lamina propria. IBS patients who report near-immediate symptom onset following specific foods may be experiencing exactly this TRM cell-mediated response, not a psychosomatic or motility-only phenomenon.

2.2 Non-Celiac Gluten Sensitivity: The ATI-Dominant Middle

Non-celiac gluten sensitivity (NCGS) — defined by gastrointestinal and systemic symptoms triggered by gluten-containing foods in the absence of celiac autoimmunity (negative tTG-IgA, EMA-IgA) and wheat allergy — affects an estimated 6% of the population, substantially exceeding celiac disease prevalence. Its pathophysiology has been progressively clarified: the primary immunological driver appears to be the ATI fraction of wheat rather than gliadin, with ATI-mediated TLR4 activation on monocytes, macrophages, and dendritic cells producing innate immune activation — without the adaptive T-cell and autoantibody responses that define celiac disease.

The gastrointestinal manifestations of NCGS — bloating, abdominal pain, altered bowel habit, and urgency — are produced by this TLR4-driven innate intestinal immune activation and by the secondary effects of increased intestinal permeability that gliadin-mediated zonulin release produces concurrently. The systemic manifestations — fatigue, brain fog, headache, joint pain, and skin changes including rash and urticaria — reflect the systemic distribution of the ATI-driven cytokine output through the mesenteric lymphatic and portal circulation, consistent with the SGIC mechanism.

Within the SGIC framework, Neu5Gc xenosialitis at the gut epithelium constitutes an independent amplifier of the NCGS presentation: anti-Neu5Gc antibody-mediated complement activation at the mucosal surface produces additional NF- κ B activation, elevates the same TNF- α , IL-6, and IL-1 β that ATI-driven TLR4 signalling produces, and independently damages tight junction proteins. An NCGS patient who simultaneously consumes bovine dairy and red meat is therefore maintaining two independent sources of intestinal inflammation and permeability that are mechanistically additive and potentially synergistic.

2.3 Celiac Disease: The HLA-Restricted Adaptive Apex

Celiac disease — affecting approximately 1% of the global population — is the best-characterised gluten-related disorder. Its pathogenesis is driven by HLA-DQ2 or DQ8-restricted CD4+ T-cell responses to deamidated gliadin peptides presented by antigen-presenting cells in

the intestinal lamina propria. The resulting adaptive immune cascade produces tissue transglutaminase (tTG) autoantibodies, endomysial antibody (EMA) production, and progressive villous atrophy that impairs intestinal absorptive function, producing malabsorption of calcium, vitamin D, iron, folate, and B12.

Within the SGIC framework, Neu5Gc xenosialitis occupies a specific and important role in celiac disease pathogenesis: by independently damaging intestinal tight junction proteins and increasing mucosal permeability, Neu5Gc-driven xenosialitis pre-conditions the mucosal environment in a way that facilitates gliadin peptide access to subepithelial antigen-presenting cells. In individuals who carry HLA-DQ2 or DQ8 but have not yet developed celiac disease — an estimated 30–40% of the general population — the Neu5Gc-mediated permeability amplification may represent one of the environmental factors that determines whether the genetic risk is expressed clinically.

The well-documented epidemiological correlation between celiac disease and dairy consumption deserves specific attention in this context. Large cohort studies have identified associations between high dairy intake and celiac disease risk that are not fully explained by shared dietary patterns or confounders. Within the SGIC framework, this association has a specific mechanistic explanation: dairy-derived Neu5Gc damages gut barrier integrity through xenosialitis-driven tight junction disruption, enhancing gliadin peptide translocation into the lamina propria in HLA-susceptible individuals and potentially facilitating the initiation of the adaptive immune response that develops into celiac disease.

2.4 Dermatitis Herpetiformis: The Cutaneous-Gastrointestinal Bridge

Dermatitis herpetiformis (DH) is the cutaneous manifestation of celiac disease — an intensely pruritic, vesicular skin condition characterised by IgA immune complex deposition at dermal papillae, driven by the same gliadin-reactive immune response that produces villous atrophy in the intestinal mucosa. DH represents the clearest example in the SGIC framework of the gut-skin axis: a gastrointestinal immune response to dietary antigen producing a cutaneous manifestation through immune complex distribution.

Within the SGIC framework, Neu5Gc xenosialitis amplifies DH through two mechanisms. First, xenosialitis-driven increases in intestinal permeability enhance the systemic availability of gliadin-reactive immune complexes — the IgA complexes that deposit at dermal papillae and produce the characteristic DH rash. Greater intestinal permeability means greater systemic immune complex burden and, potentially, more severe cutaneous IgA deposition. Second, xenosialitis at the dermal endothelium — through the same NF- κ B-driven endothelial activation mechanism documented in the Hypertension and Dyslipidaemia white paper — increases dermal vascular permeability, facilitating IgA complex deposition at the dermal-epidermal junction.

The Acnerase application's gluten/gliadin/ATI toggle — designed to identify wheat-derived antigens in foods, beverages, and medications — is directly applicable to DH management: any exposure to gliadin, regardless of source, can trigger DH lesions, and hidden gliadin sources in processed foods, medications, and supplements are a common cause of persistent DH in patients who believe they are following a strict gluten-free diet.

3. Neu5Gc's Three Roles in the Gastrointestinal Branch

3.1 Independent Tight Junction Damage

Dietary Neu5Gc incorporated into gut epithelial glycoproteins is engaged by circulating anti-Neu5Gc IgG antibodies, generating immune complexes that activate the classical complement pathway and NF- κ B at the mucosal surface. This produces measurable degradation of tight junction proteins ZO-1, occludin, and claudin-1 — the same proteins that gliadin-mediated zonulin release disrupts — through a mechanism that is entirely independent of wheat consumption. An individual consuming bovine dairy and red meat without any wheat is still maintaining Neu5Gc-driven gut barrier disruption. An individual consuming both is maintaining two independent sources of tight junction damage simultaneously, producing a degree of intestinal hyperpermeability that exceeds what either antigen alone would generate.

3.2 Pharmacokinetic Amplification of Gliadin Absorption

As established in the Hypertension and Dyslipidaemia white paper in this series, the permeability that Neu5Gc xenosialitis creates at the intestinal epithelium provides an enhanced paracellular route for the absorption of dietary antigens, including gliadin-derived peptides. Under conditions of xenosialitis-driven tight junction disruption, gliadin peptides gain access to the paracellular route — direct passage between enterocytes into the subepithelial space — without the transcellular processing that might partially attenuate their immunological activity. The effective gliadin antigenic load presented to subepithelial immune cells is therefore greater in an individual who simultaneously consumes Neu5Gc sources than it would be from gliadin consumption alone.

This pharmacokinetic interaction explains a clinical observation that is familiar to many NCGS and celiac patients but poorly understood: why symptoms following wheat consumption are more severe on some occasions than others, and why the same quantity of wheat can produce variable responses depending on concurrent dietary composition. A meal containing both wheat and bovine dairy — which describes the majority of Western meals — produces a greater gliadin antigenic load at the subepithelial immune surface than an equivalent wheat-only meal, because the Neu5Gc from the dairy component is amplifying the intestinal permeability that determines how much gliadin gets through.

3.3 Mucosal Immune Priming and the TRM Cell Gastrointestinal Model

Repeated dietary antigen exposure seeds the intestinal lamina propria with tissue-resident memory T cells (TRM cells) bearing T cell receptors specific for Neu5Gc-glycopeptide and gliadin epitopes. Upon re-exposure — whether through wheat consumption, bovine dairy consumption, or both simultaneously — these TRM cells respond within minutes of epithelial antigen contact, producing immediate cytokine release that drives rapid epithelial permeability changes, fluid secretion, and enteric nervous system activation.

This GI TRM cell model — first proposed in the context of the Soup Flare observation documented in the Arthritis/Soup Flare sentinel case report — explains the rapid-onset gastrointestinal symptoms that many NCGS, IBS, and celiac patients report: the near-immediate bloating, urgency, or pain following trigger food consumption that is inconsistent with classical food intolerance kinetics but directly consistent with a sensitised gut lamina propria responding through pre-positioned memory immune cells rather than through the slower kinetics of naive T cell priming.

The clinical corollary is significant: food challenge protocols designed to test for celiac disease or NCGS that assume a 1–4 hour symptom latency window will systematically underestimate both the speed and the severity of the gastrointestinal immune response in fully primed individuals. A patient whose bloating begins at 15 minutes rather than 2 hours is not having a different type of reaction — they are having a TRM cell-mediated reaction rather than a classical food intolerance reaction, and the distinction has diagnostic and mechanistic importance.

4. The Clinical Spectrum Table

Condition	Primary Antigen Driver	Immune Mechanism	Gut Barrier Status	Systemic Manifestations	Neu5Gc Role
IBS-D / functional GI sensitivity	ATIs (primary); Neu5Gc (amplifier)	TLR4-driven innate activation; GI TRM cell rapid response	Mildly increased permeability; subclinical	Fatigue; brain fog; extraintestinal symptoms in subset	Independent NF-κB/TJ damage; amplifies ATI response
Non-Celiac Gluten Sensitivity	ATIs (primary); gliadin (secondary); Neu5Gc (amplifier)	TLR4 innate + gliadin-mediated permeability; no adaptive autoimmunity	Moderately increased; FABP2 and zonulin elevated	Fatigue; brain fog; joint pain; skin changes; headache	Compounds TJ disruption; enhances gliadin paracellular absorption; amplifies systemic cytokine output
Celiac Disease	Gliadin (primary, HLA-restricted); ATIs (cofactor); Neu5Gc (amplifier)	HLA-DQ2/DQ8 CD4+ T-cell; tTG autoantibody; villous atrophy	Severely disrupted; malabsorption of Ca, VitD, Fe, folate, B12	Anaemia; osteoporosis; infertility; neurological; cutaneous (DH)	Pre-conditions mucosal permeability enabling gliadin translocation; may facilitate initiation of adaptive response in HLA-susceptible individuals
Dermatitis Herpetiformis	Gliadin IgA immune complexes; Neu5Gc (amplifier)	IgA immune complex deposition at dermal papillae; TG3 autoantibodies	Celiac-equivalent gut disruption	Intense pruritic vesicular rash at extensor surfaces; concurrent celiac	Amplifies IgA complex systemic burden through permeability; increases dermal vascular permeability for IgA deposition

Table 1. The Gastrointestinal Antigen Reactivity Spectrum. Each condition shares the same upstream antigen sources and intestinal permeability mechanism. Severity and clinical expression are determined by the degree of HLA-restricted adaptive immune involvement and the combined Neu5Gc plus gliadin/ATI antigenic burden.

5. Why Gluten-Free Is Often Not Enough

A significant proportion of celiac disease patients who adopt a strict gluten-free diet continue to experience gastrointestinal symptoms, fatigue, and systemic inflammation. The conventional explanation focuses on gluten contamination — hidden gluten in processed foods, cross-contamination in food preparation, or non-compliance. These explanations account for some cases. They do not account for all.

Within the SGIC framework, an additional explanation is mechanistically compelling: patients who adopt a strict gluten-free diet but continue to consume bovine dairy and red meat are maintaining Neu5Gc xenosialitis-driven tight junction disruption and intestinal mucosal NF-κB activation independently of any gluten exposure. Their gut barrier remains compromised. Their mucosal inflammatory tone remains elevated. And their capacity for intestinal recovery from the villous atrophy of celiac disease is being limited by an ongoing inflammatory stimulus that the gluten-free diet has not addressed.

For these patients, the question that has not been asked is: what else, beyond gluten, is driving the intestinal permeability and mucosal inflammation that is preventing full recovery? The SGIC framework proposes a specific answer: dietary Neu5Gc from the bovine dairy and red meat that remain in the diet. And the Acnerase application provides the practical tool to identify and eliminate the hidden sources of that antigen that make complete avoidance difficult without specialised knowledge.

Similarly, NCGS patients who follow a gluten-free diet but continue to consume bovine dairy may be maintaining the Neu5Gc-driven component of their intestinal inflammatory burden and experiencing only partial symptom resolution as a result. The severity hierarchy established in the Hamer Loop Manuscript Series predicts exactly this: remove one antigen, get partial relief; complete elimination of both antigens is necessary for complete resolution.

6. The Acnerase Gluten/ATI Toggle: Practical Application

The Acnerase food-label scanning application was designed with a specific toggle for gliadin and ATI detection, developed in recognition that the gut-skin connection documented in the Hamer Loop framework makes wheat-derived antigens directly relevant to the management of cutaneous inflammatory conditions. This toggle has expanded utility across all conditions in the gastrointestinal branch of the SGIC.

For celiac disease and DH management, the toggle identifies hidden sources of wheat-derived gliadins and ATIs in processed foods, beverages, medications, and supplements — including the non-obvious sources that are common causes of inadvertent gluten exposure in patients who believe they are following a strict gluten-free diet:

- Medications using wheat starch as a tablet excipient or wheat-derived processing aids
- Supplements using gluten-containing fillers or anti-caking agents
- Processed foods labelled ‘gluten-free’ but containing ATI fractions that are technically gluten-free but immunologically active
- Fermented and hydrolysed wheat products whose gliadin content is below the standard gluten threshold but whose ATI content remains immunologically significant
- Cross-contaminated products from shared production facilities

For NCGS and IBS management, the toggle identifies the ATI burden of foods independently of their gliadin content — important because ATI-driven symptoms can occur with foods that pass standard gluten thresholds and that patients would not expect to be problematic. The combined Neu5Gc and ATI/gliadin identification provided by the full Acnerase screening logic allows NCGS and IBS patients to address both arms of their gastrointestinal antigen burden simultaneously, rather than addressing only the gluten arm and leaving the Neu5Gc arm active.

7. The Elimination Protocol: A Staged Approach

For patients with conditions across the gastrointestinal spectrum, a staged dietary elimination approach is proposed that reflects the severity hierarchy established in the Hamer Loop Manuscript Series and the pharmacokinetic interaction between the two antigens:

Stage	Action	Expected Outcome	Timeline
Stage 1	Remove all ammonium-potentiased processed foods (caramel colour III/IV, natural flavours in savoury products, enriched flour). Use Acnerase to identify hidden sources.	Reduction in most severe trigger exposures; measurable symptom reduction in IBS and NCGS within 1–2 weeks	Days 1–14
Stage 2	Remove all concentrated bovine dairy fats (butter, ghee, aged cheese, AMF, cream). Retain whole-food dairy temporarily for palatability.	Further reduction in Neu5Gc xenosialitis-driven gut barrier damage; improved tight junction integrity; partial permeability restoration	Days 7–21
Stage 3	Remove all wheat-containing products (including hidden ATI	ATI-driven TLR4 innate activation ceases; gliadin-mediated zonulin release	Days 14–28

Stage	Action	Expected Outcome	Timeline
	sources identified by Acnerase toggle). Retain non-bovine animal proteins.	stops; tight junction restoration accelerates	
Stage 4	Remove all remaining bovine dairy (whole milk, yoghurt, soft cheese) and red meat. Complete dual-antigen elimination.	Complete xenosialitis clearance; full tight junction restoration; TRM cell quiescence in absence of cognate antigen; full symptom resolution target	Days 21–42
Maintenance	Sustain complete elimination; use Acnerase for ongoing label scanning; monitor for re-exposure sensitivity amplification.	Stable symptom resolution; ongoing protection from inadvertent antigen re-exposure; gradual reduction in anti-Neu5Gc antibody titres	Ongoing

Table 2. Staged Elimination Protocol for Gastrointestinal SGIC Branch Conditions. The staged approach reduces the behavioural burden of simultaneous elimination while systematically removing antigen sources in order of severity impact. Complete dual-antigen elimination in Stage 4 is necessary for full symptom resolution.

Note on re-exposure sensitivity: as documented in Section 6.5 of Hamer Loop Document 1, complete clearance of systemic antigen load produces a marked amplification of re-exposure sensitivity. Gastrointestinal re-exposure responses in the post-elimination state may be faster in onset and more severe than the same exposure would have produced in the chronically loaded state. Patients should be counselled about this phenomenon before beginning elimination, to prevent the misinterpretation of a post-elimination re-exposure response as evidence that the elimination protocol has made them ‘more sensitive’ to the food.

8. Testable Predictions

Prediction	Investigation Design	Expected Finding
Anti-Neu5Gc antibody titres will be elevated in celiac, NCGS, and IBS-D patients relative to matched controls and will correlate with intestinal permeability markers	Cross-sectional study: serum anti-Neu5Gc IgG; urinary lactulose:mannitol ratio; serum FABP2 and zonulin; dietary Neu5Gc burden by food frequency questionnaire; stratification by condition (celiac/NCGS/IBS/control)	Progressive elevation of anti-Neu5Gc titres and permeability markers across the spectrum from control to IBS to NCGS to celiac; correlation between Neu5Gc dietary burden and permeability measures

Prediction	Investigation Design	Expected Finding
Concurrent dairy elimination will produce greater symptom reduction in celiac patients on a gluten-free diet than gluten-free diet alone	Randomised controlled trial in celiac patients already on strict gluten-free diet: dairy elimination arm vs continued gluten-free diet without dairy change; primary: gastrointestinal symptom severity score and serum inflammatory markers at 12 weeks; secondary: mucosal healing on biopsy at 12 months	Greater symptom reduction, lower hsCRP and TNF- α , and faster mucosal healing in dairy elimination arm; consistent with Neu5Gc as independent driver of persistent mucosal inflammation in gluten-free celiac patients
Rapid-onset gastrointestinal symptoms in NCGS and IBS-D patients will be associated with elevated GI TRM cell markers and will precede the systemic cytokine elevation predicted by classical food intolerance models	Time-series blood sampling and serum cytokine measurement at 0, 15, 40, 90, and 180 minutes following controlled wheat-containing meal challenge; correlation of symptom onset time with GI TRM cell activation markers (IFN- γ , TNF- α from peripheral blood)	Symptom onset at 15–40 minutes in subset of NCGS/IBS-D patients; serum cytokine elevation preceding or coincident with symptom onset; inconsistent with 1–4 hour classical food intolerance latency
Dual elimination (wheat + bovine dairy) will produce greater and more sustained symptom improvement in NCGS than wheat elimination alone	Crossover trial: wheat elimination alone vs wheat + dairy elimination; washout period; primary: gastrointestinal symptom severity score (GSRS) at 8 and 16 weeks; secondary: anti-Neu5Gc IgG, intestinal permeability markers, serum inflammatory markers	Superior symptom reduction in dual elimination arm; correlation between residual Neu5Gc burden and residual symptom severity; pharmacokinetic interaction hypothesis confirmed if dual arm shows disproportionate permeability improvement
DH lesion burden will correlate with concurrent Neu5Gc dietary intake independent of gliadin intake	Cross-sectional study in DH patients on gluten-free diet; DH lesion severity score; dietary Neu5Gc burden by food frequency questionnaire; serum anti-Neu5Gc IgG; serum IgA anti-tTG and anti-DGP	Higher DH lesion severity in higher Neu5Gc dietary burden subgroup despite equivalent gluten-free diet adherence; correlation between anti-Neu5Gc titres and IgA complex systemic burden

Table 3. Testable Predictions. Prediction 2 is the most immediately actionable for the existing celiac patient population: it tests dual elimination in patients already on gluten-free diets who continue to have symptoms — a group for whom current management has nothing further to offer.

9. Limitations

This paper presents a structured hypothesis. The specific claim that Neu5Gc xenosialitis independently drives tight junction disruption in the human gut at concentrations achieved through habitual dietary consumption has mechanistic support from in vitro studies (He et al., 2023) but has not been directly measured in biopsied intestinal tissue from dietary consumers versus non-consumers in a human population study. The quantitative contribution of Neu5Gc versus gliadin to overall intestinal permeability in conditions across the gastrointestinal spectrum has not been directly measured.

The GI TRM cell hypothesis — proposed to explain the rapid-onset gastrointestinal symptoms of some NCGS and IBS-D patients — is mechanistically coherent and structurally homologous to the established cutaneous TRM cell model in the Hamer Loop framework. Direct evidence for Neu5Gc-specific and gliadin-specific TRM cells in human intestinal lamina propria has not been demonstrated by biopsy and immunophenotyping in dietary consumers. The hypothesis is supported by the temporal evidence of the Soup Flare observation but requires direct cellular confirmation.

The proposed role of Neu5Gc in facilitating celiac disease initiation in HLA-susceptible non-celiac individuals is the most speculative claim in this paper. The epidemiological correlation between dairy consumption and celiac disease risk has been documented but its mechanistic interpretation as Neu5Gc-driven is a novel proposition. Longitudinal cohort studies tracking Neu5Gc dietary burden, anti-Neu5Gc titres, and celiac disease incidence in HLA-DQ2/DQ8-positive individuals would be required to test this claim.

The staged elimination protocol proposed in Section 7 is based on the severity hierarchy established in the Hamer Loop Manuscript Series and applied to the gastrointestinal context. Its efficacy has not been formally tested in a randomised controlled trial in any of the gastrointestinal conditions addressed in this paper.

10. Conclusion

The gut does not lie. It is the first immunological responder to dietary antigens, the most immediate site of the consequences of intestinal permeability dysfunction, and the most direct readout of what the dietary antigen burden is doing to the mucosal immune system. The gastrointestinal symptoms of celiac disease, NCGS, IBS, and DH are not mysterious functional disturbances without identifiable cause. They are predictable immune responses to specific antigenic triggers — and those triggers are not limited to wheat.

For the patient on a strict gluten-free diet who continues to have symptoms, the SGIC framework asks a question that current management does not: what else are you eating that is driving the mucosal inflammation and barrier disruption that your gluten-free diet alone cannot resolve? For the IBS patient who has been told their condition is functional and stress-related, the framework asks: have you been tested for the two dietary antigens that are most capable of producing rapid-onset gastrointestinal immune responses in a sensitised gut? For the DH patient who cannot understand why their skin continues to flare despite strict gluten avoidance, the framework asks: are you consuming the bovine dairy that is amplifying the systemic IgA

immune complex burden and increasing the vascular permeability that deposits those complexes in your skin?

These are not rhetorical questions. They are testable hypotheses with specific, measurable predictions. The investigation they call for is within reach of existing gastroenterological research infrastructure. And the patients who need it — who are managing conditions that gluten-free diet alone has not resolved, who have been told there is nothing further to offer — are waiting for someone to ask.

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